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# The Fama-French Asset Pricing Models: Emerging Markets

Master's Thesis

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**Abstract**

The purpose of this thesis is to evaluate the performance of the Fama-French Three-factor, Five-factor and Six-factor model using stock market returns from the emerging markets. The sample has been retrieved from the Kenneth R. French Data Library and contains data from 26 countries covering July 1992 to July 2021. The size and investment factor are found to be redundant. Analysis of model performance indicates that the Three-factor model produces slightly more significant results while the Five-factor model is superior in explaining and predicting the average returns. The Six-factor model manifests an explanatory power similar or greater to the Five-factor model and heavily outperforms both the Three-factor and Five-factor model in explaining the returns of portfolios sorted on momentum, however, overall the Six-factor model's results are highly insignificant. The value and profitability factor are the primary drivers of asset returns. As a whole, the Five factor model is concluded to be a viable alternative to the Three-factor model while the Six-factor model's results are insignificant.

**Keywords:** Fama-French asset pricing models, Kenneth R. French Data Library, Emerging markets, Jensen's alpha, GRS F-test.

# Contents

**1 Introduction** **3**

**2 Theory** **5**
2.1 The Capital Asset Pricing Model 5
2.2 The Fama-French Three-factor Model 6
2.3 The Fama-French Five-factor Model 6
2.4 The Fama-French Six-factor Model 7

**3 Data** **8**

**4 Methodology** **8**
4.1 Definitions 8
4.2 Portfolio and Factor Construction 8
4.3 Test Portfolios 10
4.4 Factor Spanning Regressions 10
4.5 Jensen's Alpha 10
4.6 GRS F-test 10
4.7 Regression Intercepts 11
4.8 Empirical Models 11
4.9 Strengths and Weaknesses 11

**5 Previous studies** **12**
5.1 "A five-factor asset pricing model" - Fama and French (2015) 12
5.2 "International tests of a five-factor asset pricing model" - Fama and French (2017) 12
5.3 "Choosing factors" - Fama and French (2018) 13
5.4 "Size, value, and momentum in emerging market stock returns" - Cakici, Fabozzi and Tan (2013) 13
5.5 "Do the size, value, and momentum factors drive stock returns in emerging markets?" - Cakici, Tang and Yan (2016) 13
5.6 "Noisy prices and the Fama–French Five-factor asset pricing model in China" - Lin (2017) 14
5.7 "Size, value, profitability, and investment: Evidence from emerging markets" - Leite, Klotzle, Pinto and Silva (2018) 14
5.8 "A comprehensive test of the Fama-French Three-factor model in emerging markets" - Foye (2018) 15
5.9 "Testing factor models in Indonesia" - Foye and Valentinčič (2020) 15
5.10 "The Fama-French five-factor model: Evidence from the Johannesburg Stock Exchange" - Cox and Britten (2019) 15
5.11 "Application of asset pricing models: evidence from Saudi exchange" - Salameh (2020) 16

**6 Results** **16**
6.1 Factor Summary Statistics 16
Table 1: Factor descriptive statistics. 16
Table 2: Factor correlations. 17
6.2 Factor Spanning Regressions 17
Table 3: Factor spanning regressions. 17
6.3 GRS F-test 18
Table 4: GRS F-test results. 18
6.4 Regression Intercepts 20
Table 5: Intercepts for Size - Book-to-Market portfolios (2x3). 20
Table 6: Intercepts for Size - Operating Profitability portfolios (2x3). 21
Table 7: Intercepts for Size - Investment portfolios (2x3). 22
Table 8: Intercepts for Size - Momentum portfolios (2x3). 23
Table 9: Intercepts for Book-to-Market - Operating Profitability portfolios (2x2). 24
Table 10: Intercepts for Book-to-Market - Investment portfolios (2x2). 25
Table 11: Intercepts for Operating Profitability - Investment portfolios (2x2). 26

**7 Discussion** **27**

**References** **30**

# 1 Introduction

The purpose of this thesis is to evaluate and compare the performance of the Fama-French Three-factor, Five-factor and Six-factor model in the emerging markets. The study is novel as it is the first study to present an extensive analysis of the Six-factor model on a broader sample in the emerging markets. Furthermore, it is also, to my knowledge, the first study to analyze the 26 country emerging markets sample from Kenneth R. French's Data Library. The sample covers a comparatively large period, July 1992 to July 2021, and includes data from five continental regions: Asia, Africa, Eastern Europe, Latin America and the Middle East.

Financial economists and investors have for decades attempted to use asset pricing models to explain and predict the returns on the stock market. In the mid 1960's the Capital Asset Pricing Model (CAPM) was developed. The CAPM describes an asset's return as a function of market risk and was at first successful. However, with increased research it became clear that the CAPM was unable to explain certain patterns of returns, especially patterns related to non-market risk.

In response, Eugene Fama and Kenneth R. French, two prominent researchers in financial economics, published the Three-factor model (Fama, French 1993). The Three-factor model augments the CAPM with a size (SMB) and value (HML) factor, which represent two previously discovered anomalies known as the size and value effect. The size effect reflects the tendency of small sized stocks to outperform big sized stocks, and the value effect reflects the tendency of high book-to-market ratio stocks to outperform low book-to-market ratio stocks. The model was successful as it consistently outperformed the CAPM.

Nevertheless, researchers discovered that the Three-factor model, similar to the CAPM, had difficulty explaining several other patterns of returns. This led to the development of the Five-factor (Fama, French 2015) and the Six-factor model (Fama, French 2018) which augment the Three-factor model with the profitability (RMW), investment (CMA) and momentum (WML) factors.

Since the introduction of the Fama-French asset pricing models, there has been extensive analysis of the stock market returns in the developed markets. In general the Five-factor model and Six-factor model have shown promising results as they tend to outperform the Three-factor model in the developed markets, except in Japan (Fama, French 2015, 2017, 2018). However, fewer studies have analyzed the stock market returns in the emerging markets.

The emerging markets are capital markets that in the last decades have grown significantly. According to the investment bank Morgan Stanley (2021), the emerging markets share of global market capitalization has expanded from 19% in 2009 to 26% in 2021, indicating an increased importance with regards to global economic activity. Researchers have noted that the emerging markets have different characteristics compared to developed markets, and are often not as well integrated with each other (Cakici et al 2013). Democratic or non-democratic institutions, standards for financial reporting (Foye, Valentincic 2020), unstable political environments (Leite et al 2018) and other differences in socio-cultural behavior and norms influencing economic activity (Hammami 2020), may affect the decision making and patterns of returns observed in these markets.

Previous studies in the emerging markets have found conflicting results with regards to the performance of the Fama-French asset pricing models. Lin (2017) found the Five-factor model to consistently outperform the Three-factor model in China. Leite et al (2018) and Foye (2018), both studying a broader sample, found inconsistent results in Asia. Leite et al (2018) found that the Five-factor model outperforms the Three-factor model while Foye (2018) found that the Five-factor model does not improve on the Three-factor model. Foye and Valentincic (2020) concluded that these conflicting results primarily are due to differences in sample breadth, and therefore chose to analyze Indonesia, a larger capital market which was not included in Leite et al's (2018) sample. Their results show, similar to Foye (2018), that the Five-factor model at best offers a negligible improvement over the Three-factor model. Foye and Valentincic (2020) attribute these results to an idiosyncratic financial reporting environment characterized by insufficient accounting and auditing.

Currently, no other studies have presented an extensive analysis of the Six-factor model on a broader sample in the emerging markets. Foye and Valentincic (2020) analyzed the Six-factor model in Indonesia and found that the Six-factor model does not improve on the Five-factor model. Cakici et al (2013) analyzed the momentum factor using the Carhart model (Three-factor model + momentum factor) and found that it often is outperformed by the Three-factor model.

Sample wise, the main difference compared to other previous studies is that this thesis's sample is not regionally clustered. Additionally, compared to Foye (2018) and Cakici et al's (2013) larger eighteen country samples, it includes an additional eight countries (Qatar, Saudi Arabia, United Arab Emirates, Peru, Greece, South Africa, Egypt and Pakistan). By far, the two largest capital markets added to this thesis's sample are South Africa (1.3 trillion dollars in 2021) and Saudi Arabia (400 billion dollars in 2018 before the addition of Saudi Aramco). Due to the size of their capital markets, these two capital markets should have the largest impact on the returns of a value-weighted portfolio. Therefore two studies by Cox and Britten (South Africa, 2019) and Salameh (Saudi Arabia, 2020), both comparing the Three-factor and Five-factor model, have been included in my analysis (see *Previous studies 5.10-11)*. The results presented in both studies indicate that the Five-factor model performs similarly to the Three-factor model; the Six-factor model is not included in their analysis.

The methodology applied in this thesis is consistent with other previous studies. Model performance, i.e explanatory power, is primarily evaluated using Jensen's alpha based statistics. Jensen's alpha describes the average difference in realized asset returns and those predicted by an asset pricing model. The Gibbons-Ross-Shanken F-test (1989) is based on Jensen's alpha and is used to determine if the alphas for a set of portfolios are jointly indistinguishable from zero (see *4.3 GRS F-test*). In total, seven sets of test portfolios sorted on different factor combinations are used to determine the explanatory power of the asset pricing models. All test portfolios are value-weighted, meaning that all assets included in a portfolio are weighted according to their market value relative to the aggregate market value of all assets in the portfolio (Gruber et al, 2014).

The results indicate that the Five-factor model is a viable alternative to the Three-factor model in the emerging markets. Though the Three-factor model's results are slightly more significant, the Five-factor model often exhibits an equal or superior explanatory power. The Six-factor model heavily outperforms the Three-factor and Five-factor model in explaining the returns of portfolios sorted on momentum and, for other portfolio sets, manifests an explanatory power equal or greater than the Five-factor model. However, a majority of the Six-factor model's results are insignificant and therefore it is concluded that the model is outperformed by the Five-factor model.

The disposition is as follows. Section 2 describes the Capital Asset Pricing Model (CAPM) and the Fama-French asset pricing models. Section 3 presents the data. Section 4, *Methodology,* details the factor and portfolio construction as well as the empirical methods used to evaluate model performance. Section 5 describes the methodology, data and conclusions from eleven previous studies in both the emerging and developed markets. Due to the methodological details presented in section 5, which are explained in section 4 *Methodology*, I have chosen to place the previous studies section after the methodology. Section 6 presents the results and section 7 a discussion regarding the results and future research.

# 2 Theory

## 2.1 The Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is one of the first and most well-known asset pricing models and was independently developed by Sharpe (1964), Lintner (1965) and Mossin (1966). The CAPM is a single factor model as it only takes into account the excess market return. The coefficient of this factor, the beta, measures the comovement between an assets return and the market return (Gruber et al, 2014).

$$E(R_i) = R_F + B_i\,(E(R_M) - R_F)\text{ , where } B_i = \frac{cov(R_i, R_M)}{var(R_M)}.$$

$E(R_i)$ equals the expected return for the asset. $R_F$ equals the risk-free return. $E(R_M) - R_F$ equals the expected market return minus the risk-free return, also called the market premium. $B_i$ measures the covariance between an asset's return and the market return divided by the variance of the market return, also called market risk (Gruber et al, 2014).

The theoretical framework underlying the CAPM was built on Harry Markowitz (1959) mean-variance portfolio theory. Markowitz's idea was that all assets, to different degrees, are correlated with each other. He thought that risk could be divided into two parts: market risk and non-market risk. Non-market risk, which is risk that is unique for a set of firms with specific characteristics, could be eliminated by investing in assets that have correlation coefficients smaller than one with each other, i.e diversification. This would yield a portfolio with a return equal to the average return of the assets in the portfolio, meanwhile achieving a lower total risk than the average risk of the assets included in the portfolio. According to Markowitz, this means that all investors own a part of the market portfolio; a portfolio with the highest returns per unit of risk, also called the efficient frontier, and the rest is invested in the risk-free rate. The allocation of capital into the market portfolio and the risk-free rate, depends on the investors risk preferences. (Gruber et al, 2014).

Fundamentally, the CAPM describes the relationship between an asset's expected return and its exposure to market risk. With an increased exposure to market risk, i.e a higher beta, an investor can achieve a higher expected return. In other words, a beta above one means that an asset is more volatile than the market and a beta below one means that an asset is less volatile than the market.

Though the CAPM has been praised for its successes as the first asset pricing model to clearly define the relationship between risk and return, the model has also been the subject of criticism. The CAPM is built on several assumptions; a number of them considered unrealistic. For example, the CAPM assumes that all investors are rational, that they can borrow money at a risk-free rate and that there are no transaction costs. Moreover, many researchers question if market risk, which the beta measures, is the form of risk that is relevant, non-market risk may also have an impact on asset pricing. (Gruber et al, 2014).

Increased empirical research has given support to this criticism as several pricing anomalies have been discovered. Anomalies are patterns of asset returns that deviate from the predicted returns of an asset pricing model. There is no consensus concerning how many anomalies there are as it depends on which model is used as a benchmark and how statistically robust each anomaly is regarded to be. Hou, Mo, Xue and Zhang (2021) list 150 anomalies that they regard as statistically significant, and are difficult to explain with current models. These 150 anomalies are divided into six categories: momentum, value-versus-growth, investment, profitability, intangibles, and trading frictions.

Explanations for different types of anomalies are primarily divided into two categories: risk-based and behavioral. Risk-based explanations relate pricing anomalies to financial risk, for example credit risk or liquidity risk, and are based on the assumption that markets are efficient. This view implies that the risk-adjusted returns for different assets are the same. The behavioral explanations instead relate pricing anomalies to behavioral biases; this perspective does not rely on the market being efficient (Yang, Li & Hsu, 2017) (Gruber et

al, 2014). Anomalies that have risk-based explanations are more commonly used as factors in asset pricing models as they tend to have a more extensive and clear, though often weak, theoretical framework as well as more robust empirical evidence.

## 2.2 The Fama-French Three-factor Model

In 1993 Fama and French introduced the Three-factor model, which is a multi-factor model. The model was developed in response to the anomalies that had been discovered when testing the CAPM. The Three-factor model adds a *market capitalization* factor (SMB) as well as a *book-to-market ratio* factor (HML) to the CAPM. These two factors are based on the size effect and the value effect.

Empirical evidence for the size effect had been discovered by Banz (1981), who observed that small sized companies have higher risk-adjusted returns than big sized companies. Fama and French (1988), Chan, Hamao, and Lakonishok (1991) and Lakonishok, Shleifer, and Vishny (1993) found that companies with a high book-to-market ratio had higher risk-adjusted returns than companies with low book-to-market ratios.

In addition to the empirical evidence, Fama and French (1993) motivate the inclusion of the size and value factor by arguing that both factors are related to economic fundamentals. They had previously shown (Fama, French 1992b) that firms that have a high book-to-market ratio tend to have lower earnings and vice versa. Similarly, when controlled for book-to-market ratio, they found that size is negatively related to earnings. Fama and French conclude that both the size and value effects are related to a common risk-factor, namely relative profitability. Another factor effect that at the time had been discussed was the earnings-to-price ratio effect. Basu (1977) found that the CAPM could not price companies with high earnings-to-price ratios (E/P) as they had higher risk-adjusted excess returns. However, Fama and French (1989) concluded that the E/P effect largely disappears once the size and value factor are added.

Overall, the addition of these two factors increased the explanatory power from 70% to 90% compared to the CAPM (Fama, French 1993).

## 2.3 The Fama-French Five-factor Model

Fama and French (2015) augment the Three-factor model by adding an *operating profitability* and *investment* factor, thereby creating the Five-factor model. The profitability factor reflects the tendency of high profitability stocks to outperform low profitability stocks, and the investment factor reflects the tendency of low investment stocks to outperform high investment stocks. The theoretical framework for the profitability and investment factor is based on the dividend discount model as well as theory from Miller and Modigliani (2016).

According to the dividend discount model an assets market value is equal to the sum of its discounted dividends. $m_t$ equals the price of an asset in time t, $E(d_{t+\tau})$ equals the expected dividend of an asset in period $t + \tau$ and $r$ is the long-term average expected stock return, also called the internal rate of return on expected dividends. Equation (1) below implies that if two assets have different prices but the same expected dividends, then the asset with a lower price will have a higher expected return. If the price of an asset is considered rational, then the future dividends of the asset with a lower price must have a higher risk (Fama, French 2015).

$$m_t = \sum_{t=1}^{\infty} E(d_{t+\tau})/(1 + r)^t \qquad (1)$$

By manipulating the first equation with equation (2), Fama and French derive the relationship between the market value of an asset, the expected profitability, expected investment and book-to-market ratio. Miller and Modigliani (1961) show that for time $t$ an assets total market value can be described by the following equation (Fama, French 2015).

$$M_t = \sum_{t=1}^{\infty} E(Y_{t+\tau} - dB_{t+\tau})/(1 + r)^t \qquad (2)$$

$Y_{t+\tau}$ equals the total equity earnings in period $t + \tau$ and $dB_{t+\tau} = B_{t+\tau} - B_{t+\tau-1}$ equals the change in total book equity. By dividing the equation (2) with book equity ($B_t$), Fama and French (2015) derive the third equation with which they come to three theoretical conclusions.

$$\frac{M_t}{B_t} = \frac{\sum_{t=1}^{\infty} E(Y_{t+\tau} - dB_{t+\tau})/(1+r)^t}{B_t} \qquad (3)$$

1. Higher book-to-market equity ratio implies a higher expected return (value factor).
2. Higher expected earnings implies a higher expected return (profitability factor).
3. Higher expected growth in book equity implies a lower expected return (investment factor).

The first theoretical conclusion had already been captured in the Three-factor model. However, Fama and French (2015) discovered that the value factor becomes redundant once the profitability and investment factor are added. They argue that the value factor, due to market capitalization being sensitive to forecasts of earnings and investment, may be a "noisy proxy" for expected returns. The second conclusion is supported by Novy-Marx (2013), who found empirical evidence for a relationship between average returns and profitability. The last conclusion is supported by Aharoni, Grundy and Zeng (2013) who found a connection between the investment and average returns.

Fama and French (2015) conclude that the Five-factor model improves on the Three-factor model. However, their study focuses on stock returns in the United States and they therefore encourage further analysis of other regions that may have different characteristics.

## 2.4 The Fama-French Six-factor Model

In 2018 Fama and French decided to add the *momentum* factor to the Five-factor model. The momentum factor was first added to the Three-factor model by Carhart in 1997 and reflects the tendency of stocks that have had above average returns in past periods to continue to do well in future periods, and vice versa. Fama and French define a stock's momentum as its cumulative returns between months t-12 to t-2. Risk-based explanations for the momentum effect have been difficult to verify as the effect appears to be unrelated to other risk-factors. Researchers have instead commonly linked the momentum effect to behavioral biases, however, no consensus has been reached. Moreover, Fama and French (2018) have criticized behavioral explanations and argue that behavioral biases should not persist over longer periods of time as they expect investors to eventually recognize and arbitrage such market inefficiencies.

Multi-factor models such as the Three-, Five- and Six-factor model are, as previously mentioned, a response to the empirical failures of the CAPM. Yet, according to Fama and French (2018) a positive aspect with the CAPM is that it applies a clear theoretical framework to specify the relationship between expected returns and risk. Multi-factor models are, on the other hand, mainly driven by patterns observed in historical data. However, new patterns are discovered yearly and therefore Fama and French (2018) argue that there is a risk that the development of new factor models becomes an exercise in data dredging as many anomalies, though statistically robust, lack theoretical support. If the factors are not related to economic fundamentals, it may also lead to less effective asset pricing models as it decreases the likelihood that the asset pricing model's performance is robust across markets with different characteristics.

Nonetheless, the main reason for adding the momentum factor to the Five-factor model was the continued use of the Carhart model as well as there being extensive evidence that neither the Three-factor or Five-factor model could adequately price momentum (Fama, French. 2012, 2016b, 2018).

# 3 Data

The sample has been retrieved from Kenneth R. French's Data Library in September 2021 and includes data from 26 countries in the emerging markets. It contains monthly factor returns for all six factors used in the Six-factor model as well as monthly returns for 36 different portfolios used for both factor construction and as test portfolios. The following countries are included: Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Malaysia, Mexico, Pakistan, Peru, Philippines, Poland, Qatar, Russia, Saudi Arabia, South Africa, South Korea, Taiwan, Thailand, Turkey and the United Arab Emirates. The original sample period is 1989 to 2021, however, due to variation in start dates for different portfolio and factor returns as well as missing values, the sample period used in this thesis will be July 1992 to July 2021 (349 observations). The returns include dividends and capital gains, are measured in U.S. dollars and are not continuously compounded.

# 4 Methodology

The methodology applied in this thesis is primarily based on Fama and French (2015), Leite et al (2018) and Foye (2018). This methodology is considered standard and is consistently used by prominent researchers in both the developed and emerging markets. Moreover, applying a similar methodology simplifies the comparison of results.

## 4.1 Definitions

**Market capitalization**, also called market equity, is calculated as a stocks closing price times shares outstanding. (Kenneth R.French Data Library, September 2021)

**Book equity** equals the book value of stockholders' equity, plus deferred taxes and investment tax credits, minus the book value of preferred stock. (Kenneth R.French Data Library, September 2021)

**Size** is defined as the market capitalization of a firm. (Kenneth R.French Data Library, September 2021)

**Value** is defined as a firm's book-to-market equity ratio (B/M). (Kenneth R.French Data Library, September 2021)

**Operating profitability** is defined as annual revenues minus cost of goods sold, interest expense, and selling, general, and administrative expense, all divided by the sum of book equity and minority interest for the last fiscal year ending in t-1. (Kenneth R.French Data Library, September 2021)

**Investment** is defined as the change in total assets from the fiscal year ending in year t-2 to the fiscal year ending in t-1, divided by total assets in t-2. (Kenneth R.French Data Library, September 2021)

**Momentum** is defined as a stock's cumulative return for month t–12 to month t–2. (Kenneth R.French Data Library, September 2021)

## 4.2 Portfolio and Factor Construction

*The portfolios and factors used in this thesis as well as information detailing the factor and portfolio construction has been retrieved from the Kenneth R.French Data Library. Predominantly, a Bloomberg terminal has been used to construct the portfolios.*

**The market factor**, $R_{Mt} - R_{Ft}$, equals the return on a region's value-weighted market portfolio minus the U.S. one month Treasury bill rate.

**The size (SMB), value (HML), operating profitability (RMW) and investment (CMA) factors** have been constructed using the three 2x3 value-weighted portfolio sets sorted on size-B/M, size-profitability and

size-investment (see subsection *4.3 Test portfolios*). Stocks are sorted into two market capitalization and three B/M, operating profitability and investment categories at the end of June each year.

For the size factor, big stocks are those in the top 90% of June market capitalization for the country, and small stocks are those in the bottom 10%. The B/M, operating profitability, and investment breakpoints for a country are the 30th and 70th percentiles of respective ratios for the big stocks in the country.

**SMB (Small Minus Big)** equals the average return on the nine small stock portfolios minus the average return on the nine big stock portfolios.

| | | |
|---|---|---|
| $SMB_{(B/M)}$ | = | 1/3 (Small Value + Small Neutral + Small Growth) - 1/3 (Big Value + Big Neutral + Big Growth) |
| $SMB_{(OP)}$ | = | 1/3 (Small Robust + Small Neutral + Small Weak) - 1/3 (Big Robust + Big Neutral + Big Weak). |
| $SMB_{(INV)}$ | = | 1/3 (Small Conservative + Small Neutral + Small Aggressive) - 1/3 (Big Conservative + Big Neutral + Big Aggressive) |
| SMB | = | 1/3 ( $SMB_{(B/M)}$ + $SMB_{(OP)}$ + $SMB_{(INV)}$ ) |

**HML (High Minus Low)** equals the average return on the two value portfolios minus the average return on the two growth portfolios.

| | | |
|---|---|---|
| HML | = | 1/2 (Small Value + Big Value) - 1/2 (Small Growth + Big Growth) |

**RMW (Robust Minus Weak)** equals the average return on the two robust operating profitability portfolios minus the average return on the two weak operating profitability portfolios.

| | | |
|---|---|---|
| RMW | = | 1/2 (Small Robust + Big Robust) - 1/2 (Small Weak + Big Weak) |

**CMA (Conservative Minus Aggressive)** is the average return of the two conservative investment portfolios minus the average return on the two aggressive investment portfolios.

| | | |
|---|---|---|
| CMA | = | 1/2 (Small Conservative + Big Conservative) - 1/2 (Small Aggressive + Big Aggressive) |

The market return for July of year t to June of t+1 includes all stocks for which there is data for June of year t. SMB, HML, RMW, and CMA for July of year t to June of t+1 include all stocks for which there is data for December of t-1 and June of t, (positive) book equity data for t-1 (for SMB, HML, and RMW), non-missing revenues and one or several of the following: cost of goods sold, selling, general and administrative expenses, or interest expense for t-1 (for SMB and RMW), and total assets data for t-2 and t-1 (for SMB and CMA).

**The momentum factor** is formed monthly and is constructed using a 2x3 portfolio set sorted on size and lagged momentum. For portfolios formed at the end of month t–1, the momentum return equals a stock's cumulative return for month t–12 to month t–2. The momentum breakpoints for a country are the 30th and 70th percentiles of the momentum returns of the big stocks in each country.

**WML (Winners minus Losers)** equals the equal-weight average of the returns for the two winner portfolios for emerging markets minus the average of the returns for the two loser portfolios.

| | | |
|---|---|---|
| WML | = | 1/2 (Small Winners + Big Winners) – 1/2 (Small Losers + Big Losers) |

The six portfolios used to construct WML each month include stocks with prior return data. To be included in a portfolio for month t (formed at the end of the month t–1), a stock must have a price for the end of month t–13 and a good return for t–2.

## 4.3 Test Portfolios

The test portfolios are used to evaluate the explanatory power of the Fama-french asset pricing models. Compared to the 2x3 sorted portfolio sets, the 2x2 sorted portfolio sets instead use 50th percentile factor breakpoints for all the eligible stocks of each country. Seven sets of portfolios are analyzed which in total gives 36 test portfolios.

| **All test portfolios used to evaluate model performance are value-weighted.** |
|---|
| 6 Portfolios Formed on Size and Book-to-Market (2 x 3) |
| 6 Portfolios Formed on Size and Operating Profitability (2 x 3) |
| 6 Portfolios Formed on Size and Investment (2 x 3) |
| 6 Portfolios Formed on Size and Momentum (2 x 3) |
| 4 Portfolios Formed on Book-to-Market and Operating Profitability (2 x 2) |
| 4 Portfolios Formed on Operating Profitability and Investment (2 x 2) |
| 4 Portfolios Formed on Book-to-Market and Investment (2 x 2) |

## 4.4 Factor Spanning Regressions

Factor spanning regressions are used to test for redundant factors by sequentially regressing a sixth factor on five factors. If the intercept of the factor spanning regression is significant, it indicates that the other factors do not capture the effects of the dependent factor, implying that the factor adds explanatory power to the model.

The following regression is an example of a factor spanning regression using the size factor as the dependent variable. If $a_i$ is significant, then the size factor is considered non-redundant, i.e the factor may add explanatory power to the model:

$$SMB_t = a_i + B_i(R_{Mt} - R_{Ft}) + h_i HML_t + r_i RMW_t + c_i CMA_t + u_i WML$$

## 4.5 Jensen's Alpha

Jensen's alpha is a commonly used performance measure for asset pricing models developed by Michael Jensen in 1968. It measures the average difference between the return of an asset or portfolio and the return predicted by an asset pricing model. For example, estimating the alpha for the CAPM requires four variables: the portfolio return, the market return, the risk-free rate and the coefficient between the portfolio and market return.

$$a_i = (R_{it} - R_{Ft}) - B_i(R_{Mt} - R_{Ft})$$

## 4.6 GRS F-test

The Gibbons-Ross-Shanken (1989) F-test will be used to test the null hypothesis that the alphas, estimated for a set of portfolios, are jointly indistinguishable from zero. As mentioned above, the alpha describes how well the RHS (right-hand side) factors explain the LHS (left-hand side) portfolio returns. An alpha closer to zero indicates less unexplained portfolio returns.

GRS F-test formula: $$\frac{T}{N}\frac{T-N-L}{T-L-1}\frac{\hat{a}' \cdot \widehat{\Sigma^{-1}} \cdot \hat{a}}{1+\bar{\mu}' \cdot \widehat{\Omega^{-1}} \cdot \bar{\mu}} \sim F(N, T-N-L)$$

*T* represents the number of monthly returns. *N* represents the number of test portfolios. L equals the number of factors. *F* represents the F-distribution. $\hat{a}$ is a *N x 1* vector of estimated alphas. $\widehat{\Sigma^{-1}}$ is an estimate of the residual covariance matrix of *N* error terms. $\bar{\mu}$ is a *L x 1* vector of the factor portfolios' sample means. $\widehat{\Omega^{-1}}$ is an estimate of the factor portfolios' covariance matrix without adjusting for degrees of freedom, $\hat{a}'$ is a transposed vector.

Fama and French (2018) state that $\hat{a}' \cdot \widehat{\Sigma^{-1}} \cdot \hat{a}$ equals the difference between the max squared Sharpe ratio that can be constructed from a model's factor returns and the portfolio returns, and the maximum Sharpe ratio that can be constructed with only the factor returns. The Sharpe ratio measures the average portfolio excess return per unit of volatility or risk, and is a regularly used measure of portfolio performance.

The factor models are ranked according to the size of the GRS statistic, smaller being better, as well as the p-value of the GRS statistic. For each set of portfolios, the GRS statistic will be evaluated together with the average absolute alpha and the average $R^2$-adjusted to determine the explanatory power of each model. The $R^2$ , adjusted for degrees of freedom, describes the proportion of the variation in portfolio returns (0-100%) that are explained by the model's factors.

According to Kamstra and Shi (2021) a common misrepresentation of the original GRS paper among financial economists, is the adjustment of degrees of freedom when calculating the estimator of the sample covariance matrix of the factor portfolios ($\widehat{\Omega^{-1}}$). Using algebraic proofs, simulations as well as empirical results from previous well cited studies such as Fama and French (2015), they show that the GRS statistic when adjusted for degrees of freedom causes over-rejection and often biases the results in favor of models using fewer factors.

They argue that this primarily is a problem for studies that use less than 15-20 years of data. Although the effects of using the incorrect GRS statistic tend to be small, in this thesis I choose to use the non-adjusted version of the GRS statistic to ensure that the results are reliable.

## 4.7 Regression Intercepts

Multivariate statistics as a performance measure for asset pricing models, such as the GRS F-test, were partly developed due to the difficulty summarizing the results from the univariate statistics (statistics that use one dependent variable). Therefore, in addition to the GRS F-test, the intercepts (alphas) and their significance levels, which are estimated by the LHS portfolio RHS model regressions, will be independently analyzed. This will give deeper insight into which specific cross-sections of returns that the different models have difficulty explaining.

## 4.8 Empirical Models

*Three-factor model*

$$R_{it} - R_{Ft} = a_i + B_i(R_{Mt} - R_{Ft}) + s_i SMB_t + h_i HML_t + \boldsymbol{\varepsilon}_{it}$$

*Five-factor model*

$$R_{it} - R_{Ft} = a_i + B_i(R_{Mt} - R_{Ft}) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + \boldsymbol{\varepsilon}_{it}$$

*Six-factor model*

$$R_{it} - R_{Ft} = a_i + B_i(R_{Mt} - R_{Ft}) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + u_i WML + \boldsymbol{\varepsilon}_{it}$$

If the coefficients $B_i$, $s_i$, $h_i$, $r_i$, $c_i$ and $u_i$ explain all variation in portfolio returns, then the intercept $a_i$ is zero for all portfolios $i$. $\boldsymbol{\varepsilon}_{it}$ is a zero-mean residual.

## 4.9 Strengths and Weaknesses

One of the weaknesses with my thesis is that it, in contrast to Lin (2017), Cakici et al (2013), Leite et al (2018) and Foye (2018), is not possible to analyze each country or region within the emerging markets independently. Thus, it is more difficult to understand the regional market characteristics that may affect the results. Therefore, it is important to compare the results presented in this thesis with results from other studies that have analyzed the emerging markets on a regional or country-level.

Another weakness is that there, in this sample, aren't as many portfolios available for analysis compared to other studies. This is, most likely, due to there being fewer firms in the emerging markets data which makes it difficult to more finely sort the firms without affecting the diversification of the portfolios, thereby negatively affecting the robustness of the results. Prominent studies in asset pricing often use 5x5 or 2x4x4 portfolio sorting schemes which may lead to a deeper understanding of which cross-sections of portfolio returns that different asset pricing models have difficulty explaining.

The strength of my thesis is that it uses data from a comparatively large sample period and that the sample is broad as it contains data from 26 countries, this should add robustness to the results. The countries that are included in my sample but not in broad sample studies such as Foye (2018) and Cakici et al (2013), are the following: Qatar, Saudi Arabia, United Arab Emirates, Peru, Greece, South Africa, Egypt and Pakistan. I have included two studies from South Africa and Saudi Arabia, two of the largest capital markets in the emerging markets, to my analysis (*5.10-11 Previous studies*). These two markets should have the largest impact on a value-weighted portfolio as their market capitalizations are comparatively large.

# 5 Previous studies

This section details the data, methodology, results and conclusions from eleven studies analyzing both the emerging markets and developed markets.

## 5.1 "A five-factor asset pricing model" - Fama and French (2015)

The main purpose of the study is to compare the performance of the Five-factor model to the Three-factor. Fama and French use factor spanning regressions to test for factor redundancy. Model performance is primarily evaluated with the GRS F-test and performance statistics based on Jensen's alpha. The sample covers July 1963 to December 2013. To test how sensitive the results are to different factor definitions, the factors are constructed using three different sorting schemes: 2x2, 2x3 and 2x2x2x2. The test portfolio sets are created using two different sorting schemes: 5x5 for the size-B/M, size-profitability, size-investment and 2x4x4 for the size-B/M-profitability, size-B/M-investment and size-profitability-investment portfolio sets.

The results show that the value factor becomes redundant once the profitability and investment factor are added. Fama and French (2015) argue that the value factor, due to market capitalization being sensitive to forecasts of earnings and investment, may be a "noisy proxy" for expected returns. Model performance does not seem to be affected by the factor construction method and they therefore choose to continue using the 2x3 factor construction scheme as it is commonly used in the literature. Overall, the Five-factor model outperforms the Three-factor model regardless of the factor construction method. The Five-factor model's primary problem is that it has trouble explaining the returns of small sized stocks. Especially small sized stocks with high investment and low profitability.

## 5.2 "International tests of a five-factor asset pricing model" - Fama and French (2017)

Using a similar methodology to their 2015 study on a U.S. sample, Fama and French evaluate the performance of the Five-factor model in four regions in the developed markets: North America, Europe, Japan and Asia Pacific. The main difference is that they use a shorter sample period which covers July 1990 to December 2015. The performance of the Five-factor model is compared to the performance of the Three-factor model and a Five-factor model that excludes the investment factor.

The results show that the size and investment factors are redundant in Europe and Japan. The size factor is the only redundant factor in Asia Pacific. In general, the Five-factor model outperforms the Three-factor model in all regions except Japan. In Japan, all three models produce insignificant GRS statistics for all sets of portfolios. In Europe, the main problem for the Five-factor model is explaining the returns of the size-investment sorted portfolio set. This is most likely due to the size and investment factor being redundant in that region. Similar to their study in 2015, Fama and French conclude that the primary problem of the Five-factor model is that it is not capable of explaining the returns of small stocks that have similar returns to those with low profitability and high investment.

## 5.3 "Choosing factors" - Fama and French (2018)

The purpose of the study is to analyze different versions of the Six-factor model's performance, which adds momentum to the Five-factor model. In addition, an alternative definition of the profitability factor is tested, using cash profitability instead of operating profitability. Furthermore, Fama and French test a new performance metric proposed by Barillas and Shanken (2016). This performance metric is the max squared Sharpe ratio of the intercepts from LHS factor return regressions and is mainly used to compare nested and non-nested models. The max squared Sharpe ratio is closely related to the GRS F-test, however, the GRS statistic is not suited for the comparison of non-nested models as it causes an upward bias for models that include more factors. Non-nested models are models that use distinct factors, meaning that the models do not use the same factor definitions. The sample contains data from the U.S. stock market between July 1963 to June 2016.

The factor spanning regressions indicate that the momentum factor adds explanatory power to the Five-factor model. Cash profitability is found to outperform operating profitability when analyzed using the Barillas and Shanken metric. A Six-factor model which combines the market and size factor with the small stock spread factors (meaning factors created only using small sized companies) $HML_S$, $RMW_S$, $CMA_S$, and $WML_S$ outperforms the other models with regards to the max squared Sharpe ratio statistic proposed by Barillas and Shanken. However, Fama and French conclude that this does not justify a permanent switch to these new factor definitions as the base Six-factor model also performs well. Overall, the Barillas and Shanken statistic correlates with the GRS statistic, which is not surprising as they are closely related.

## 5.4 "Size, value, and momentum in emerging market stock returns" - Cakici, Fabozzi and Tan (2013)

The size, value and momentum effects are examined in 18 emerging markets divided into three regions: Asia, Eastern Europe and Latin America. The authors use monthly stock data between January 1990 to December 2011. Factor and portfolio summary statistics as well as factor spanning regressions are used to analyze the factor effects in the emerging markets, global markets and the U.S. In addition, two sets of portfolios (5x5) sorted on size-B/M and size-momentum are analyzed using the CAPM, Three-factor model and Carhart model. The performance of the asset pricing models are also compared using factors created with local, global and U.S. data, which tests for market integration. The GRS F-test, Jensen's alpha based performance metrics as well as a GMM-based test-statistic are used to evaluate and rank the performance of the different models. GMM (Generalized Method of Moments) is used to test for non-normal and serially auto-correlated error terms. The purpose of the GMM statistic is to control the significance level of the GRS statistic.

The authors find a statistically significant value effect in all three regions in the emerging markets, with the big sized value premia being slightly larger than the small sized value premia. The reverse is found in the U.S. and global developed markets, where the small sized value premia is larger than the big sized value premia. The momentum effect is found to be significant in all regions except Eastern Europe. The momentum premia are found to be larger in small sized stocks compared to big sized stocks. This pattern of momentum premia regarding size is consistent with results found in the developed markets. Performance evaluation shows that the use of global and U.S. constructed factors decreases the explanatory power of local returns (i.e returns in different regions of the emerging markets). These results indicate that the emerging markets are not fully integrated with the developed or global markets. The Carhart model, which includes the momentum factor, is found to be comparatively successful in explaining the returns of the size-momentum sorted portfolios, especially in Asia. However, overall the momentum factor does not seem to add explanatory power. The GMM results indicate that the significance level of the GRS statistic is robust for local factors and a majority of the results using U.S. and global factors.

## 5.5 "Do the size, value, and momentum factors drive stock returns in emerging markets?" - Cakici, Tang and Yan (2016)

Cakici, Fabozzi and Tan (2016) examine the size, value and momentum effects in 18 emerging markets between 1990 to 2013. They analyze the factor effects in several different sub-periods: 1990-2002, 2002-2013, 2007-2009 (crisis) and the larger period 1990-2011. The main difference between this study and other

referenced studies from the emerging markets is that this study does not analyze the performance of any asset pricing models. The advantage of this study is that it gives deeper insight into country specific factor effects instead of regional or just the emerging markets as a whole.

They find evidence for the value effect in all emerging markets except Brazil and find the effect to be robust through all sub-periods. The momentum effect is weak and non-significant in most countries through all sub-periods and a strongly negative, albeit non-significant, momentum premium is observed during the crisis period. The size effect is not visible in most countries except China, similar results are presented by Lin (2017).

## 5.6 "Noisy prices and the Fama–French Five-factor asset pricing model in China" - Lin (2017)

The Three-factor model and Five-factor model are compared using a Chinese sample which covers July 1997 to December 2015. Factor and test portfolios are constructed using a similar methodology to Fama and French (2015). Factor spanning regressions and factor summary statistics are used to test for factor redundancy and analyze the factor premia. The GRS F-test and performance metrics based on Jensen's alpha are used to evaluate the performance of the asset pricing model models. In addition, the intercepts estimated by the LHS portfolio RHS model regressions are independently analyzed.

The value and profitability factor are found to be non-redundant, meanwhile the investment factor is found to be redundant. This is in contrast to Fama-French (2015) who instead found the value factor to be redundant when adding the investment and profitability factor. Lin (2017) argues that this is due to two reasons. Firstly, the emerging markets are usually bank oriented which means that financial institutions are the primary source of financing instead of the securities market. This leads to profitability being a better predictor, than investment, of future company performance. Secondly, companies in several emerging markets are usually characterized by a strong ownership concentration, investment may consequently predominantly be used to benefit controlling shareholders instead of the development of the company.

In all, the Five-factor model outperforms the Three-factor model in all performance metrics. However, the analysis of regression intercepts from the double-sorted portfolio sets indicate that the Five-factor model has difficulty explaining the high returns of stocks that perform like low B/M stocks that have low investment due to low profitability. Among the triple-sorted portfolio sets, the Five-factor model's primary problem is explaining the returns of portfolios with exceptionally low returns. The Six-factor model was analyzed, but no extensive results are presented (comment in footnote). The Six-factor model only outperformed the Five- and Three-factor model on the size-momentum portfolios.

## 5.7 "Size, value, profitability, and investment: Evidence from emerging markets" - Leite, Klotzle, Pinto and Silva (2018)

The sample covers January 2009 to February 2017, which is shorter than other studies. Twelve countries in the emerging markets from three different continental regions; Latin America, Eastern Europe and Asia are analyzed using the CAPM, the Three-factor, the Five-factor model - HML as well as the base Five-factor model. Model performance is measured using a similar methodology to other studies, focusing on the GRS F-test and other statistics based on Jensen's alpha. Factors are constructed and model performance compared using both 2x3 and 2x2x2x2 factor sorting schemes. In addition, model performance is measured using global and U.S. factors to analyze international market integration. Test portfolios consist of three 5x5 sets sorted on size-B/M, size-investment and size-profitability.

The results show that local factors perform better than U.S. or global factors, indicating that the emerging markets are not fully integrated with international markets. Furthermore, the Five-factor model - HML and the Five-factor model perform similarly and outperform the Three-factor model and the CAPM. Letie et al (2018) argue that this most likely is due to a redundant effect between the value, profitability and investment factor, as the Four-factor model excludes the value factor. The size effect is most prominent while the value and profitability effect are not intelligible and the investment effect is weak. The size and pattern of the factor effects

are the only results that are not similar to those found in the developed markets. Leite et al (2018) argue that this is due to political and economic problems that affect the number of liquid stocks in the stock market, which in turn leads to suboptimal diversification of portfolios.

## 5.8 "A comprehensive test of the Fama-French Three-factor model in emerging markets" - Foye (2018)

Foye (2018) uses an extensive sample of 18 countries from the emerging markets in Eastern Europe, Latin America and Asia to test the Five-factor model. The sample period stretches between July 1997 to June 2016. The methodology is comparable to Leite et al (2018), with the GRS F-test as well as Jensen's alpha metrics being the main statistics used for evaluating model performance.

The results show, similarly to Lin (2017), that there is a strong value premium in all three regions and the value factor is found to be the only factor that is not redundant in any of the regions. In addition, the results indicate that there is a profitability premium in Eastern Europe and Latin America, however, the factor is not significant in Asia. The investment factor and the size factor are both redundant in all three regions. The Five-factor model has difficulty improving on the explanatory power of the Three-factor model in Asia, this is in contrast to Leite et al (2018) and Lin (2017) who find that the Five-factor model does improve on the Three-factor model. Foye (2018) concludes that the value and profitability factor are the main sources of stock returns in Europe and Latin America, meanwhile in Asia, the value factor alone seems to be the most important source of portfolio returns.

## 5.9 "Testing factor models in Indonesia" - Foye and Valentinčič (2020)

Foye and Valentinčič (2020) study the Indonesian stock market between December 1996 and May 2017. Overall the methodology is similar to Fama and French (2015). The main difference is that the authors include the momentum factor in their analysis, however, they do not construct a set of portfolios sorted on the momentum factor. In total, sixteen models, ranging from the CAPM + SMB to the Six-factor model are evaluated and compared. The GRS F-test and performance metrics based on Jensen's alpha are used to evaluate model performance.

The momentum effect is found to be weak and statistically insignificant, overall the momentum factor does not add explanatory power to the Five-factor model. Furthermore, both the Three-factor and Five-factor model produce large intercepts for all sets of portfolios and the Five-factor model does not improve on the Three-factor model. They argue that Indonesia has properties such as an idiosyncratic financial reporting environment and a low earnings quality that contribute to these results. Studies in the emerging markets that exclude Indonesia, such as Lin (2017) and Leite et al (2018), seem to instead show that the Five-factor model outperforms the Three-factor model.

## 5.10 "The Fama-French five-factor model: Evidence from the Johannesburg Stock Exchange" - Cox and Britten (2019)

Cox and Britten study the performance of six asset pricing models, based on the factors included in the Five-factor model, using data from the Johannesburg Stock exchange between 1994 to 2017. South Africa's Johannesburg Stock exchange represents one of the biggest stock exchanges in the emerging markets with a market cap of 1.28 trillion dollars as of October 2021 (Johannesburg Stock Exchange). South Africa is included in the sample used in this thesis but not in the other studies.

The factor summary statistics show that the market and investment premium are weak and insignificant, meanwhile the size premium is strongly negative and significant. The value and profitability factor are both large and significant. Analysis of model performance indicates that the Three-factor model and a size-profitability Three-factor model slightly outperform the Five-factor model in explaining time-series (GRS F-test) returns of a majority of the portfolio sets. On the other hand, the Five-factor model outperforms the other models in explaining the cross-sectional returns. The profitability factor consistently adds explanatory power while the investment factor does not. The weakness of the study is that the authors do not report the p-value of

the GRS statistic. Additionally, 10% is used as the lowest threshold for a significant result, which is high and questions the significance of their results.

### 5.11 "Application of asset pricing models: evidence from Saudi exchange" - Salameh (2020)

Salameh compares the performance of the CAPM, the Three-factor and Five-factor model using data from Saudi Arabia's Stock Exchange (Tadawul). The Saudi Arabian stock exchange is one of the largest in the emerging markets and had a market cap of 4-500 billion dollars in 2018, one year before the inclusion of Saudi Aramco, which increased the market capitalization to 2.5 trillion dollars (December 2021). Saudi Arabia is not included in any other referenced studies in the emerging markets (see Foye's (2018) sample).

Salameh concludes that all three models perform similarly, with the Five-factor model producing a slightly higher adjusted $R^2$, but a majority of the results are insignificant. It is primarily the market and size factor that drive the results, the investment and profitability factor do not seem to add any explanatory power. Salameh argues that it has been difficult to apply asset pricing models in Saudi Arabia due to Islamic Sharia (borrowing restrictions, no leverage and no risk-free asset) which makes it difficult to identify the correct factors. The weakness of the study is that it uses a comparatively short sample period, January 2014 to August 2017, and the threshold for a significant result is set at 10%.

## 6 Results

### 6.1 Factor Summary Statistics

Table 1: Factor descriptive statistics.

Mean (%) equals the average returns of the factor portfolio, also known as factor premium or effect, T-statistic > 1.96 indicates that the factor premium is significantly different from zero. Sample period: July 1992 - July 2021.

| Factors -> | Market | SMB | HML | RMW | CMA | WML |
|---|---|---|---|---|---|---|
| **Mean (%)** | 0.67 | 0.11 | 0.61 | 0.18 | 0.24 | 0.83 |
| **SD (%)** | 6.10 | 2.11 | 2.30 | 1.61 | 1.97 | 2.96 |
| **T-statistic** | 2.05 | 0.96 | 4.94 | 2.08 | 2.36 | 5.22 |

Table 1 shows that the factor premiums and their significance levels are rather similar to Fama and French (2018), except that the size factor's premium is insignificant. Average returns for the size factor are 0.11% but non-significant (0.96), which is consistent with Cakici et al (2016), Foye (2018) and Foye & Valenticic (2020) who also found the size factor to be insignificant. Lin (2017) reports the size factor to be the only significant factor effect in China. A highly significant and large value premium with average monthly returns of 0.61% (4.94) can be observed. Foye (2018) and Cakici et al (2013) found similar results in Asia, meanwhile Lin (2017) found the value factor to be redundant in China and Leite et al (2018) found all factor premiums to be insignificant in all regions.

Additionally, a weakly significant but small profitability and investment premium can be observed. Foye & Valenticic (2020) found a statistically significant negative investment and positive profitability premium in Indonesia. The profitability premium seems to be driven by Latin America and Eastern Europe (Foye, 2018). The investment premium is somewhat peculiar. Foye and Valentincic (2020) found the investment premium to be significant but negative in Indonesia. Other studies such as Foye (2018), Leite et al (2018) and Lin (2017) also found the investment premium to be insignificant and oftentimes negative. There is a highly significant and large momentum premium of 0.83% (5.22) which is consistent with Cakici et al (2013). The market premium is large and significant. Cakici et al (2013) found the market factor to be insignificant and sometimes negative.

Table 2: Factor correlations.

| Factors | Market | SMB | HML | RMW | CMA | WML |
|---|---|---|---|---|---|---|
| **Market** | 1.000 | -0.208 | 0.111 | -0.292 | -0.294 | -0.217 |
| **SMB** | -0.208 | 1.000 | 0.015 | -0.163 | 0.044 | 0.141 |
| **HML** | 0.111 | 0.015 | 1.000 | -0.514 | 0.355 | -0.167 |
| **RMW** | -0.292 | -0.163 | -0.514 | 1.000 | -0.140 | 0.198 |
| **CMA** | -0.294 | 0.044 | 0.355 | -0.140 | 1.000 | 0.109 |
| **WML** | -0.217 | 0.141 | -0.167 | 0.198 | 0.109 | 1.000 |

Among all the factor correlations (Table 2) the value factor and the profitability factor have the highest negative correlation at -0.514. The value and investment factor have the highest positive factor correlation at 0.355. This is expected as high B/M stocks tend to have lower profitability and higher investment (Fama-French, 1993, 2015). The market factor has a negative correlation with all factors except the value factor (0.111), between -0.2 to -0.3.

The size and value factor show a minimal correlation of 0.015. Foye (2018) found a strong correlation between the size factor and the investment factor. In this sample, the size factor and the investment factor instead show a weak correlation of 0.044. Fama and French (2015), found a positive correlation between the size and market factor (0.28). In this sample the correlation is negative (-0.208), which indicates that small stocks may have lower betas.

The value and momentum factors negative correlation (-0.167) is not surprising as similar, but often stronger, negative correlations are presented by Cakici et al (2013, 2016) in the emerging markets and Asness et al (2013) in both the developed and emerging markets. Though there is no consensus regarding the explanation for this negative correlation, Asness et al (2013) have shown that value and momentum have the opposite relationship to liquidity risk. In general, momentum increases while liquidity is available. Value or high B/M firms, on the other hand, tend to instead increase their returns during liquidity shocks. This pattern is especially visible during periods of financial crisis, for example the Great Recession between 2007-2009.

## 6.2 Factor Spanning Regressions

Table 3: Factor spanning regressions.

A sixth factor is sequentially regressed on five factors. The table presents the intercept, coefficients and adjusted $R^2$ for each regression. A non-significant intercept indicates that the dependent factor is redundant. *** = 0.1% significance level, ** = 1% significance level, * = 5% significance level.

| LHS \| RHS | Market | SMB | HML | RMW | CMA | WML | Intercept | Adj $R^2$ |
|---|---|---|---|---|---|---|---|---|
| **Market** | | -0.691 | 0.130 | -1.29 | -1.074 | -0.147 | 1.282*** | 0.259 |
| **SMB** | -0.099 | | -0.062 | -0.418 | -0.083 | 0.099 | 0.226 | 0.114 |
| **HML** | 0.016 | -0.054 | | -0.637 | 0.377 | -0.075 | 0.687*** | 0.353 |
| **RMW** | -0.076 | -0.174 | -0.306 | | -0.059 | 0.056 | 0.404*** | 0.367 |
| **CMA** | -0.110 | -0.060 | 0.313 | -0.102 | | 0.079 | 0.087 | 0.245 |
| **WML** | -0.040 | 0.200 | -0.175 | 0.269 | 0.223 | | 0.837*** | 0.094 |

The results from the factor spanning regressions (Table 3) show that the intercepts for the excess market return (Market), value (HML), profitability (RMW) and the momentum factor (WML) are all highly significant. It can

consequently be concluded that the value factor, in contrast to Fama and French (2015) but similar to Cakici et al (2013), Lin (2017) and Foye (2018), is not redundant.

Lin (2017) and Foye (2018) found the investment factor and the size factor to be redundant in Asia. We can likewise conclude that the investment factor is redundant as its intercept is insignificant. The size factor is also insignificant and redundant. In the developed markets, Fama and French (2017) found the size and investment factor to be redundant in Europe and Japan. Cakici et al (2013) and Leite et al (2018) do not present any factor spanning regressions.

## 6.3 GRS F-test

Table 4: GRS F-test results.

*** = 0.1% significance level, ** = 1% significance level, * = 5% significance level, GRS=GRS statistic, p(GRS)=p-value GRS statistic, alpha=average alpha for all portfolios in the portfolio set, |alpha|=average absolute alpha for all portfolios in the portfolio set, Adjusted $R^2$=Average adjusted $R^2$ for portfolios in the portfolio set. A smaller GRS statistic and lower average absolute alpha indicates higher explanatory power. Sample period: July 1992 - July 2021.

| | | | | |
|---|---|---|---|---|
| ***Panel A: Size - Book/Market (2x3)*** | | | | |
| **Models** | **GRS** | **p(GRS)** | **\|alpha\|** | **Adjusted $R^2$** |
| **FF3M** | 4.162 | 0.000*** | 0.148 | 0.971 |
| **FF5M** | 2.919 | 0.009** | 0.142 | 0.972 |
| **FF6M** | 1.936 | 0.074 | 0.119 | 0.976 |
| ***Panel B: Size - Operating profitability (2x3)*** | | | | |
| **Models** | **GRS** | **p(GRS)** | **\|alpha\|** | **Adjusted $R^2$** |
| **FF3M** | 9.345 | 0.000*** | 0.198 | 0.964 |
| **FF5M** | 3.398 | 0.003** | 0.092 | 0.976 |
| **FF6M** | 3.604 | 0.002** | 0.095 | 0.982 |
| ***Panel C: Size - Investment (2x3)*** | | | | |
| **Models** | **GRS** | **p(GRS)** | **\|alpha\|** | **Adjusted $R^2$** |
| **FF3M** | 1.505 | 0.176 | 0.078 | 0.974 |
| **FF5M** | 1.568 | 0.156 | 0.065 | 0.982 |
| **FF6M** | 1.867 | 0.086 | 0.065 | 0.982 |
| ***Panel D: Size - Momentum (2x3)*** | | | | |
| **Models** | **GRS** | **p(GRS)** | **\|alpha\|** | **Adjusted $R^2$** |
| **FF3M** | 11.916 | 0.000*** | 0.382 | 0.939 |
| **FF5M** | 10.192 | 0.000*** | 0.340 | 0.940 |
| **FF6M** | 5.785 | 0.000*** | 0.084 | 0.971 |
| ***Panel E: Book/Market - Operating Profitability (2x2)*** | | | | |
| **Models** | **GRS** | **p(GRS)** | **\|alpha\|** | **Adjusted $R^2$** |
| **FF3M** | 9.230 | 0.000*** | 0.180 | 0.963 |
| **FF5M** | 3.370 | 0.0106* | 0.095 | 0.967 |
| **FF6M** | 2.096 | 0.081 | 0.092 | 0.968 |

*Panel F: Book/Market - Investment (2x2)*

| Models | GRS | p(GRS) | \|alpha\| | Adjusted $R^2$ |
|---|---|---|---|---|
| **FF3M** | 2.851 | 0.024* | 0.093 | 0.960 |
| **FF5M** | 2.399 | 0.050* | 0.092 | 0.967 |
| **FF6M** | 0.961 | 0.429 | 0.065 | 0.968 |

*Panel G: Operating Profitability - Investment (2x2)*

| Models | GRS | p(GRS) | \|alpha\| | Adjusted $R^2$ |
|---|---|---|---|---|
| **FF3M** | 7.092 | 0.000*** | 0.170 | 0.960 |
| **FF5M** | 1.668 | 0.157 | 0.067 | 0.969 |
| **FF6M** | 2.035 | 0.089 | 0.083 | 0.968 |

*Summary of the GRS F-test results*

| Models | Number of significant GRS statistics | Number of significant alphas |
|---|---|---|
| **FF3M** | 6/7 | 21/36 |
| **FF5M** | 5/7 | 17/36 |
| **FF6M** | 2/7 | 9/36 |

The results from panel A *(Size - Book/Market)* indicate that both the Three-factor and Five-factor model perform well. Both models produce a highly significant GRS statistic and similar average absolute alphas (0.140-0.150), however, the Five-factor model's GRS statistic is smaller. The Six-factor model produces a lower GRS statistic as well as average absolute alpha but the results are non-significant.

Panel B *(Size - Operating Profitability)* shows that the Five-factor and Six-factor model heavily outperform the Three-factor model. All three GRS results are highly significant. The Five-factor model slightly outperforms the Six-factor model.

In panel C *(Size - Investment)*, all three models produce insignificant results. This is most likely due to the investment and size factor being redundant and the portfolio set being sorted on these two factors. Similar results can be observed in Lin (2017) and Foye (2018) but not in Foye & Valenticic (2020). It seems as if it is the Asian countries except Indonesia that drive these results.

The panel D *(Size - Momentum)* results show that all three models produce highly significant results. However the Six-factor model produces considerably smaller GRS statistic as well as an average absolute alpha of 0.084, which is significantly lower than the Five-factor model and Six-Factor model at 0.340 and 0.382 respectively. We can conclude that the Three-factor and Five-Factor model, both have difficulty explaining the returns of portfolios sorted on momentum.

Panel E *(Book/Market - Operating Profitability)* shows that the Five-factor model performs better than the two other models. It produces a low GRS statistic (3.37) and average absolute alpha (0.095), heavily outperforming the Three-factor model. The Six-factor model's results are insignificant.

Panel F *(Book/Market - Investment)* reveals that the Three-factor and Five-factor model perform similarly. The Three-factor model's results are slightly more significant. The Six-factor model's results are highly insignificant at 42.9%.

Panel G *(Operating Profitability - Investment)* shows that both the Five-factor and Six-factor model outperform the Three-factor model in explaining the returns of the profitability-investment portfolio set, however, the results are insignificant.

## 6.4 Regression Intercepts

Table 5: Intercepts for Size - Book-to-Market portfolios (2x3).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **Size \| B/M** | **Low** | **Mid** | **High** |
|---|---|---|---|
| **Small** | 0.200 | 0.714 | 1.142 |
| **Big** | 0.569 | 0.699 | 0.841 |

***Panel A: FF3 Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| B/M** | **Low** | **Mid** | **High** | **Size \| B/M** | **Low** | **Mid** | **High** |
| **Small** | **-0.285** | -0.040 | **0.120** | **Small** | **-3.97** | -0.66 | **2.49** |
| **Big** | **0.197** | -0.040 | **-0.209** | **Big** | **3.68** | -0.69 | **-3.20** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| B/M** | **Low** | **Mid** | **High** | **Size \| B/M** | **Low** | **Mid** | **High** |
| **Small** | **-0.249** | -0.012 | **0.139** | **Small** | **-3.34** | -0.18 | **2.65** |
| **Big** | **0.190** | -0.064 | **-0.198** | **Big** | **3.23** | -0.97 | **-2.70** |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| B/M** | **Low** | **Mid** | **High** | **Size \| B/M** | **Low** | **Mid** | **High** |
| **Small** | **-0.200** | 0.035 | **0.105** | **Small** | **-2.46** | 0.48 | **1.98** |
| **Big** | **0.159** | -0.067 | -0.145 | **Big** | **2.70** | -0.95 | -1.75 |

The portfolio average returns in Table 5 show that there is a value premium in both size categories. Similar to Fama and French (2015) the value premium is stronger for small sized companies. Cakici et al (2013) instead found the big sized value premium to be stronger than the small sized value premium. We do not observe a size premium in the low B/M category.

All three models produce intercepts close to zero for the small and big sized mid B/M portfolios, however, the intercepts are non-significant. The Six-factor model produces the lowest intercepts for all four of the other portfolios, however, the big size high B/M portfolio intercept is not significant (t-statistic = -1.75). Overall the Three-factor model and Five-factor model produce four significant intercepts. Compared to the Three-factor model, the Five-factor Model produces slightly smaller intercepts for three out of four of the significant intercepts.

Table 6: Intercepts for Size - Operating Profitability portfolios (2x3).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **Size \| RMW** | **Low** | **Mid** | **High** |
|---|---|---|---|
| **Small** | 0.711 | 0.992 | 0.817 |
| **Big** | 0.518 | 0.638 | 0.773 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| RMW** | **Low** | **Mid** | **High** | **Size \| RMW** | **Low** | **Mid** | **High** |
| **Small** | **-0.209** | **0.194** | 0.102 | **Small** | **-4.05** | **4.16** | 1.47 |
| **Big** | **-0.314** | -0.088 | **0.278** | **Big** | **-4.90** | -1.71 | **5.21** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| RMW** | **Low** | **Mid** | **High** | **Size \| RMW** | **Low** | **Mid** | **High** |
| **Small** | -0.001 | **0.143** | **-0.176** | **Small** | -0.03 | **2.68** | **-2.74** |
| **Big** | -0.062 | -0.056 | **0.113** | **Big** | -1.05 | -1.02 | **2.14** |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| RMW** | **Low** | **Mid** | **High** | **Size \| RMW** | **Low** | **Mid** | **High** |
| **Small** | -0.024 | **0.143** | **-0.181** | **Small** | -0.61 | **2.51** | **-2.56** |
| **Big** | -0.044 | -0.065 | **0.114** | **Big** | -0.70 | -1.10 | **2.00** |

The average returns of the size-profitability sorted portfolios indicate that it primarily is the returns of the big sized portfolios that drives the profitability premium observed in the summary statistics in Table 1. There is no profitability premium between small sized mid and high profitability portfolios. A size premium can be observed in all profitability categories.

The Three-factor model produces four significant intercepts, meanwhile the Five-factor and Six-factor model produces three significant intercepts. The Five- and Six-factor model manifest a similar pattern in that they both produce insignificant intercepts for the small and big sized low profitability and big sized mid profitability portfolios. Compared to the Five- and Six-factor model, the Three-factor model produces significant intercepts for the small and big sized low profitability portfolios. Overall the Five- and Six-factor model produce smaller significant intercepts than the Three-factor model.

Table 7: Intercepts for Size - Investment portfolios (2x3).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns (%).***

| **Size \| CMA** | **Low** | **Mid** | **High** |
|---|---|---|---|
| **Small** | 0.935 | 0.957 | 0.588 |
| **Big** | 0.748 | 0.684 | 0.607 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| CMA** | **Low** | **Mid** | **High** | **Size \| CMA** | **Low** | **Mid** | **High** |
| **Small** | 0.049 | 0.114 | **-0.175** | **Small** | 0.80 | 1.95 | **-2.11** |
| **Big** | 0.050 | -0.041 | 0.036 | **Big** | 0.60 | -0.78 | 0.49 |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| CMA** | **Low** | **Mid** | **High** | **Size \| CMA** | **Low** | **Mid** | **High** |
| **Small** | 0.048 | 0.118 | -0.061 | **Small** | 0.92 | 1.84 | -0.82 |
| **Big** | -0.015 | -0.056 | 0.095 | **Big** | -0.23 | -0.96 | 1.59 |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| CMA** | **Low** | **Mid** | **High** | **Size \| CMA** | **Low** | **Mid** | **High** |
| **Small** | 0.010 | **0.161** | -0.078 | **Small** | 0.18 | **2.29** | -0.99 |
| **Big** | -0.006 | -0.055 | 0.082 | **Big** | -0.09 | -0.90 | 1.26 |

The pattern of the average portfolio returns indicate that there is an investment premium, except between the low and mid investment levels for small sized portfolios.

All three models have difficulty producing significant intercepts for this set of portfolios. This is most likely due to the size and investment factor both being redundant. Foye (2018) found a similar pattern.

Table 8: Intercepts for Size - Momentum portfolios (2x3).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **Size \| WML** | **Low** | **Mid** | **High** |
|---|---|---|---|
| **Small** | 0.342 | 1.097 | 1.259 |
| **Big** | 0.277 | 0.656 | 1.013 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| WML** | **Low** | **Mid** | **High** | **Size \| WML** | **Low** | **Mid** | **High** |
| **Small** | **-0.533** | **0.298** | **0.484** | **Small** | **-5.09** | **5.48** | **4.88** |
| **Big** | **-0.514** | -0.030 | **0.439** | **Big** | **-5.00** | -0.58 | **4.81** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| WML** | **Low** | **Mid** | **High** | **Size \| WML** | **Low** | **Mid** | **High** |
| **Small** | **-0.416** | **0.311** | **0.393** | **Small** | **-3.31** | **5.34** | **3.70** |
| **Big** | **-0.485** | -0.054 | **0.379** | **Big** | **-4.15** | -1.04 | **3.88** |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | | **t(alpha)** | | |
|---|---|---|---|---|---|---|---|
| **Size \| WML** | **Low** | **Mid** | **High** | **Size \| WML** | **Low** | **Mid** | **High** |
| **Small** | 0.072 | **0.318** | 0.026 | **Small** | 0.78 | **5.17** | 0.31 |
| **Big** | -0.044 | -0.042 | 0.001 | **Big** | -0.57 | -0.77 | 0.01 |

The results in Table 8 indicate that there is a strong momentum premium in both size categories and between all momentum categories.

All three models produce a large intercept for the small size mid momentum portfolio and a small intercept for the big size mid momentum portfolio. Regarding significance level, a small minority of the Six-factor model's intercepts are significant (1/6) while a large majority of the Three-factor and Five-factor model's intercepts are significant (5/6).

Table 9: Intercepts for Book-to-Market - Operating Profitability portfolios (2x2).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **B/M \| RMW** | **Low** | **High** |
|---|---|---|
| **Low** | 0.537 | 0.621 |
| **High** | 0.874 | 0.683 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| RMW** | **Low** | **High** | **B/M \| RMW** | **Low** | **High** |
| **Low** | -0.131 | **0.211** | **Low** | -1.50 | **3.96** |
| **High** | **-0.206** | **0.171** | **High** | **-4.49** | **2.03** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| RMW** | **Low** | **High** | **B/M \| RMW** | **Low** | **High** |
| **Low** | 0.046 | **0.101** | **Low** | 0.55 | **1.98** |
| **High** | **-0.149** | 0.085 | **High** | **-3.17** | 0.93 |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| RMW** | **Low** | **High** | **B/M \| RMW** | **Low** | **High** |
| **Low** | -0.082 | 0.081 | **Low** | -1.00 | 1.48 |
| **High** | -0.078 | 0.125 | **High** | -1.57 | 1.29 |

In Table 9 we continue to observe a value premium. The profitability premium is only visible between the low B/M portfolios.

The Six-factor model is unable to produce any significant intercepts. The Three-factor model produces three significant intercepts but similar to the Five-factor model it has difficulty producing a significant intercept for the low B/M low profitability portfolio. Both models produce significant intercepts for the high B/M low profitability and low B/M high profitability portfolios but the Five-factor model produces significantly smaller intercepts.

Table 10: Intercepts for Book-to-Market - Investment portfolios (2x2).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **B/M \| CMA** | **Low** | **High** |
|---|---|---|
| **Low** | 0.402 | 0.659 |
| **High** | 0.710 | 0.961 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| CMA** | **Low** | **High** | **B/M \| CMA** | **Low** | **High** |
| **Low** | 0.029 | **0.143** | **Low** | 0.34 | **1.98** |
| **High** | -0.005 | **-0.194** | **High** | -0.09 | **-2.95** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| CMA** | **Low** | **High** | **B/M \| CMA** | **Low** | **High** |
| **Low** | 0.014 | **0.146** | **Low** | 0.20 | **2.48** |
| **High** | -0.022 | **-0.184** | **High** | -0.34 | **-2.63** |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **B/M \| CMA** | **Low** | **High** | **B/M \| CMA** | **Low** | **High** |
| **Low** | -0.047 | 0.106 | **Low** | -0.62 | 1.69 |
| **High** | 0.002 | -0.104 | **High** | 0.04 | -1.44 |

In contrast to Table 7 (Size-Investment), in Table 10 we do not see an investment premium when analyzing the pattern of average portfolio returns. Instead we observe a strong negative investment premium.

The Three-factor and Five-factor model both produce the same two significant and equally large intercepts. Similar results are found in the developed markets (Fama, French 2015). The Six-factor model has trouble producing any significant results.

Table 11: Intercepts for Operating Profitability - Investment portfolios (2x2).

The alpha represents the intercept of each LHS portfolio RHS model regression. All portfolios are value-weighted. t(alpha) = t-statistic of intercept. Intercepts at 5% significance level are highlighted. Sample period: July 1992 - July 2021.

***Portfolio average excess returns.***

| **RMW \| CMA** | **Low** | **High** |
|---|---|---|
| **Low** | 0.710 | 0.539 |
| **High** | 0.757 | 0.750 |

***Panel A: FF3M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **RMW \| CMA** | **Low** | **High** | **RMW \| CMA** | **Low** | **High** |
| **Low** | -0.115 | **-0.205** | **Low** | -1.83 | **-3.02** |
| **High** | 0.146 | **0.216** | **High** | 1.64 | **3.35** |

***Panel B: FF5M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **RMW \| CMA** | **Low** | **High** | **RMW \| CMA** | **Low** | **High** |
| **Low** | -0.028 | -0.085 | **Low** | -0.40 | -1.38 |
| **High** | -0.018 | **0.138** | **High** | -0.23 | **2.40** |

***Panel C: FF6M Intercepts.***

| | **alpha** | | | **t(alpha)** | |
|---|---|---|---|---|---|
| **RMW \| CMA** | **Low** | **High** | **RMW \| CMA** | **Low** | **High** |
| **Low** | -0.031 | 0.106 | **Low** | -0.42 | -1.60 |
| **High** | -0.036 | **0.157** | **High** | -0.44 | **2.68** |

Table 11 shows a weak profitability premium and a small investment premium among the low profitability portfolios. The investment premium is barely noticeable in the high profitability category.

The Five-factor and Six-factor model are only capable of producing significant intercepts for the high profitability high investment portfolio. The Three-factor model produces significant intercepts for the low and high profitability high investment portfolios, and produces close to significant intercepts for the two remaining portfolios. However, the intercepts for the Three-factor model are larger compared to the the Five- and Six-factor model.

# 7 Discussion

The purpose of this thesis was to evaluate and compare the performance of the Fama-French Three-, Five- and Six-factor model in the emerging markets. The study is novel as it is the first study to analyze the 26 country emerging markets sample from Kenneth R. French's Data Library. Furthermore, it is also the first study to present an extensive analysis of the Fama-French Six-factor model on a broad sample from the emerging markets.

Overall, the factor premiums (Table 1) are rather similar to those reported in the United States (Fama, French 2018). The main difference is that the size effect is insignificant and the value effect is slightly smaller. Moreover, there is a strong and significant momentum effect in the emerging markets, Cakici et al (2013) found similar results. It is important to note that Cakici et al (2016) find the momentum effect, when analyzed at a country-level, to instead be weak and insignificant in the emerging markets.

The factor spanning regressions (Table 3) show that the value factor, in contrast to Fama and French (2015), is non-redundant. The results instead indicate that the size and investment factor are redundant. Similar results are found by Foye (2018) and Lin (2017) in Asia, but not by Foye and Valentincic (2020) who instead found the momentum factor and market factor to be redundant. In the developed markets Fama and French (2017) found the size and investment factor to be redundant in Europe and Japan.

Lin (2017) and Foye (2018) argue that there are two reasons for why the investment factor is redundant. Firstly, the emerging markets are usually bank oriented, thus financial institutions are the primary source of financing instead of the securities market. This leads to profitability being a better predictor than investment of future company performance. Secondly, companies in several emerging markets are usually characterized by strong ownership concentration, investment may consequently predominantly be used to benefit controlling shareholders instead of the development of the company. Therefore, past investment may not be a good predictor of stock returns.

Table 4 (GRS F-test) shows that the Three-factor model produces highly significant GRS statistics for six out of seven portfolio sets. The Five-factor model produces significant GRS statistics for five out of seven portfolio sets and performs especially well with regards to the portfolio sets sorted on profitability, except the investment-profitability set. In general, the Five-factor model produces lower GRS statistics and average absolute alphas than the Three-factor model.

The Six-factor model often produces a GRS statistic as well as average absolute alpha which is equal or lower than the Five-factor model's. Additionally, the Six-factor model successfully explains the returns of the size-momentum portfolio set, heavily outperforming the two other models. However, overall the Six-factor model's results are insignificant.

All three factor models produce high average adjusted $R^2$ for all sets of portfolios, usually between 96-98%, strongly correlating with the average absolute alpha and GRS statistic. The only portfolio set for which none of the models produce a significant GRS statistic is the size-investment portfolio set. This is most likely due to size and investment both being redundant factors. The value and profitability factor are the primary drivers of asset returns.

The results from the regression analysis (Table 5-11) shows that the Three-factor model produces significant intercepts for a majority of the test portfolios (21/36). The Five-factor results are not as often significant (17/36). The Six-factor model performs poorly, 9/36 intercepts are significant and only 1/6 size-momentum intercepts are significant. Though the intercepts are comparatively small, all three models consistently have difficulty producing any significant intercepts for portfolios sorted on low investment. Fama and French (2015), found that the Five-factor model has difficulty explaining small sized stocks. This pattern cannot be observed in this thesis and may partly be due to the portfolio sets not being as finely sorted.

Summarizing the results, the overall explanatory power and significance levels suggest that the Five-factor model is a viable alternative to the Three-factor model in the broader emerging markets. The Six-factor model's results are insignificant and therefore it is outperformed by the Five-factor model.

Foye and Valentincic (2020) as well as Foye (2018) found that the Five-factor and the Six-factor perform equally as well or were outperformed by the Three-factor model in Asia, with all three models producing large average absolute alphas across all portfolio sets. Both of these studies include Indonesia in their sample. In other regions such as Latin America and Eastern Europe or studies that exclude Indonesia (Leite et al, 2018) (Lin, 2017), the Five-factor model instead consistently outperforms the Three-factor model. Foye and Valentincic (2020) argue that Indonesia has properties such as an idiosyncratic financial reporting environment and a low earnings quality that contribute to these results.

The effects of the Indonesian results observed in Foye (2018) and Foye and Valentincic (2020) do not seem to have the same effect on this sample. This may be due to the large number of countries included in this sample, leading to an overall decrease in the total effect of the Indonesian results. Cox and Britten´s (2019) analysis of South Africa and Salameh's (2020) analysis of Saudi Arabia, two of the largest capital markets included in this study but not in other studies, show that the Five-factor model performs similarly to the Three-factor model. However, Cox and Britten's (2019) as well as Salameh's (2020) results are characterized by low significance levels and in Salameh's case a shorter sample period, which makes it difficult to infer the effects of those markets on this thesis's results.

To conclude, the results presented in this thesis, with regards to the Three-factor and Five-factor model, are in between those of other prominent studies in the emerging markets that had found conflicting results. Foye and Valentincic (2020) argue, in my opinion correctly, that future research should be focused on country-level data or groups of countries not only clustered on geographic proximity. Other variables that may be unique for a set of countries, such as the financial reporting environment, non-democratic institutions or socio-cultural norms and behaviors, should instead be the focal point as those variables may relay more information regarding the economic intuition behind the results.

For example, several regions in the emerging markets are Islamic: The Middle East, North Africa, Indonesia and Malaysia to name a few. Hammami (2020) characterizes Islamic economies as having borrowing restictions, i.e low or zero debt (Islam prohitibits interest rates), no leverage and no risk-free asset; all violations of the CAPM assumptions. Instead of analyzing the performance of Fama-French models, Hammami proposes a consumption based CAPM adjusted for Islamic market characteristics. As there is no risk-free asset, this model is derived using an asset whose covariance with the intertemporal marginal rate of substitution is zero (zero-covariance asset). As the results indicate that the model performs well, I suggest constructing a sample solely including Islamic markets, using a larger sample period than Salameh (2020), to compare the Fama-French models to Hammami's CCAPM.

In addition, future research in the emerging markets should analyze the performance of asset pricing models using portfolios sorted on other factor combinations or anomalies. Hou, Mo, Xue and Zhang (2021) evaluate the performance of several different asset pricing models, including the Five-factor and Six-factor model, in explaining 150 anomalies sorted into six categories. A similar analysis should be carried out in the emerging markets as this would further test the robustness of the Fama-French models and perhaps give insight into other factors or models that, from the perspective of the emerging markets, may be viable alternatives.

Hou et al (2021) also present the Q5-model, which is based on their four-factor Q-model (2012). The Q5-model has shown promising results in the U.S as it consistently outperforms the cash profitability Six-factor model in all anomalie categories. The primary difference compared to the Six-factor model, is that the Q5-model excludes the value and momentum factor. However, interestingly it appears capable of explaining the returns of portfolios sorted on momentum. The fifth factor in the Q5-model is the expected growth factor. Hou et al (2021) relate the expected growth factor to investment theory. Holding present investment and profitability constant, they argue

that firms with high expected investment should earn higher expected returns and vice versa. Therefore, if expected investment is high in the upcoming period, the present value of cash flows from the next period forth should also be high. Mainly consisting of this present value, the benefits of current investment should also be high. Thus, if current investment is lower than expected investment in the next period, the current discount rate must be high to counterbalance the high benefit of current investment to keep current investment low.

The Q5-model's exclusion of the value factor has been criticized for not being well motivated (Fama, French 2015) and may be problematic as the value factor appears to be one of the primary drivers of returns in the emerging markets. Nevertheless, the other factors, such as the expected growth factor, may compensate for the value factors exclusion. Thus giving further insight into the mechanism behind the value effect as well as other factor effects in the emerging markets.

Finally, the GRS F-test - which lies at the center of the methodologies applied in the asset pricing literature to evaluate model performance - is primarily based on two assumptions: non-serially correlated error terms and a normal distribution of returns. These two assumptions are frequently made by financial economists as they tend to simplify the mathematics underlying asset pricing theories and statistical methods of analysis. However, the first and especially the second assumption is commonly accepted by researchers to be inconsistent with the data as extreme returns tend to be more prevalent than a normal distribution would imply (Mandelbrot, 1963) (Fama, 1965).

As can be observed in other previous studies presented in section 5, a large majority of prominent researchers in asset pricing choose not to adress or test the robustness of these assumptions. Fama and French (Fama/French Forum, 2009) argue that non-normal distributions of returns do not undermine most theories and statistical tools used in asset pricing. Affleck-Graves and McDonald (1989), using non-normal distributions of returns typically found in the data, analyzed the robustness of the GRS F-test and concluded that it is fairly robust to deviations.

Cakici et al (2013), a study which is described in section 5 of this thesis, chose instead to control for these assumptions by applying a General Method of Moments (GMM) based test statistic (Zhou, 1994) which they consider robust to non-normal distributions and serially auto-correlated error terms. Their results suggest that the GRS statistic, when applied using locally constructed factors, is robust as the significance level of the GRS statistic and GMM test statistic are consistent. Future research may implement this robustness check to further increase the reliability and confidence of the results.

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